Bryce Hoenigman

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Computational linguistics student building machine learning models and data pipelines for low-resource ancient languages, open to internships in the Bay Area Summer 2027.

Technical Skills

- **Coding languages:** Python, SQL (SQLite, schema design/DDL)
- **ML & data:** PyTorch, scikit-learn, pandas, NumPy, statsmodels, Pydantic
- **Tools:** Claude Code, Git/GitHub, EMu (collections database), Praat
- **Visualization:** Matplotlib, Seaborn, Altair
- **Domain:** NLP for low-resource languages, model fine-tuning, corpus analysis, computer vision for cuneiform
- **Languages:** Arabic, Spanish, Akkadian

Education

University of Chicago — B.A., Linguistics with minors in Computational Linguistics, Data Science, and Middle Eastern Studies ◆ Expected June 2028

- **Technical coursework:** Machine Learning: Concepts and Applications; Computational Linguistics I–II; Data Science 11800–11900; Data Visualization; Computer Science 14100; AI Development Program
- **Linguistics coursework:** Phonetics and Phonology; Morphology; Semantics and Pragmatics; Intermediate Akkadian sequence, Comparative Semitics

Experience

Self-directed Cuneiform Provenience Classifier Project— Emergent Ventures Grant, Mercatus Center · 2026–Present

- Built an image classification pipeline that predicts where a cuneiform tablet was written from 158,946 sign images
- Fine-tuned ResNet101 and EfficientNetV2-S encoders on 266-class sign identification, reaching 84.3% accuracy and 0.764 macro-F1
- Trained EfficientNet-B0 provenance classifiers: 0.95 macro-F1 for north vs. south and 0.68 macro-F1 across 16 cities (baseline 0.017)
- Won the computing award at MindByte's RCC research symposium

Summer Researcher — Digital Pasts Laboratory · 2026–Present

- Analyzed Akkadian-determinative disambiguation metrics by fine-tuning a roBERT masked learning model and seq2seq translation model
- Reduced data migration friction by designing a SQL database schema to migrate data from ORACC into the iClassifier project
- Extracted determinatives from ORACC Akkadian texts in Python and ran statistical analysis to identify corpus-wide usage patterns
- Reviewed prior literature to select candidate variables for statistical modeling of determinative usage

Research Assistant, Tablet Collection — Institute for the Study of Ancient Cultures (QUAD Faculty Research Grant) · 2025–Present

- Digitized and catalogued hundreds of collection objects, efficiently processing requests from external researchers
- Transliterated, translated and line-drew tablets for the Nippur publication project
- Researched the Prisms of Ashurbanipal; presented findings at the CCRF Undergraduate Research Symposium.

Research Intern — Script Encoding Initiative, UC Berkeley · 2025–2026

- Researched Unicode zero-width joiners (ZWJ/ZWNJ) across bidirectional, Indic and emoji scripts using Stanford archival sources